ISHAN KUMARASINGHE

COMPUTER ENGINEERING UNDERGRADUATE

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PROFILE

Final year Computer Engineering undergraduate specializing in Full stack development, Cloud platforms and DevOps methodologies. Experience ranges from developing a system, configuring local environments to managing scalable, production-level deployments. Good knowledge in system administration, CI/CD pipelines, and cloud platforms. Seeking an job to apply technical skills in optimizing system reliability, performance, and scalability.

EXPERIENCE

Intern Full Stack Software Engineer **April 2026 - Present**

GTN Technologies (Pvt) Limited | Full-time - Hybrid

- Develop and maintain systems, including the EOD Report Manager, GTN-Tradelens, Release Management System, while contributing to software development and system upgrades.

EDUCATION

BSc Eng Hons specializing in Computer Engineering (UG) **Feb 2022 - Present**

University of Peradeniya

- GPA: 3.7 /4.0

TECHNICAL SKILLS

- Languages: Python, Java, C++, SQL, C, JavaScript, TypeScript, Dart, Ballerina
- Cloud: Git, GitHub, Linux, GCP, AWS, Docker, Jenkins, Ansible, Terraform
- ML frameworks: Scikit-learn, PyTorch, TensorFlow
- Development: Spring Boot, Node.js, Express.js, React, Angular, Flutter, AJAX
- Libraries & Tools: OpenCV, Trello, Keras, MLflow

PROJECTS

Synthetic Cardiac MRI Image Generation using Deep Generative Models | *Research* | link **Aug 2025 - March 2026**

Tech Stack: Python, PyTorch, MONAI, Diffusers

Synthetic Cardiac MRI Image Generation from Diffusion and Flow matching Deep Generative Models using limited publicly available data.

My contribution:

- Generate synthetic multi labeled masks to condition the deep generative models.
- Build conditional Diffusion models with 70-80 FID value to generate synthetic cardiac MRI.
- Evaluate generated synthetic images on Fidelity to measure the quality and Privacy to preserve patient privacy.

SMART AGRICULTURAL MONITORING SYSTEM | *Group* | link **Dec 2024 - Jul 2025**

Tech Stack: YOLO, FastAPI, Docker, GCP, Firebase, Platform.io, MQTT, Flutter, React

IoT solution to collect real-time location-based data, demonstrating over 30% efficiency in the field.

My contribution:

- Built a dockerized YOLOv5 tree health monitoring service using FastAPI, deployed on GCP Cloud Run with automated CI/CD pipelines for seamless updates.
- Developed a mobile and web apps fully integrated with Firebase services and Cloud Storage.
- Collected IoT sensor data via ESP32 microcontroller and stored in Firestore using MQTT.

POSTGRADUATE STUDENTS MANAGEMENT SYSTEM | *Group* | link **Apr 2024 - Nov 2024**

Tech Stack: Angular, Spring boot, MySQL, Typescript

Software solution designed to enhance administrative efficiency and academic monitoring within the department using Spring Boot, integrating JWT authentication and role-based access control.

My contribution:

- Built the Web app using Angular for Postgraduate portal introducing 6+ new features.
- Integrated the frontend with backend APIs, ensuring secure connectivity via JWT and CORS.

Event-Driven, Dockerized To-Do Application | *Individual* | link **Dec 2025 - Feb 2026**

Tech Stack: Docker Compose, React, Node.js, MySQL, RabbitMQ, AWS

A production-style full stack app with event-driven backend and containerized deployment.

- Developed asynchronous event-driven architecture using RabbitMQ achieving 100% Notification system reliability.
- Orchestrated production-ready containers via Docker Compose ensuring high availability.
- Configured cloud-ready deployment patterns for AWS.

ACHIEVEMENTS

Undergraduate Inventor of the Year (UIY) 2025 Nominee | IESL **Oct 2025**

Third-Year Project (SAMS) submitted to the competition organized by the Institution of Engineers, Sri Lanka (IESL) and ranked within the top 15 out of over 50 competing teams.

ENGEX 75th Anniversary Exhibition Showcase | University of Peradeniya **Aug 2025**

SAMS project selected for display at the SMART CITY zone by Peracom, celebrating the Faculty of Engineering's 75th anniversary and receiving positive feedbacks from industry professionals.

Innovate with Ballerina Coding Challenge | IEEE UoM & WSO2 **Nov 2024**

"Book My Doctor" project selected for the coding challenge and successfully completing the project with a high score.

CERTIFICATION

- 100 Days of DevOps - KodeKloud
- AWS Cloud Solutions Architect Professional Certificate
- Advanced Learning Algorithms - DeepLearning.AI
- AI Free Week Course - KodeKloud
- Generative AI: The Evolution of Thoughtful Online Search - LinkedIn

RELEVANT COURSEWORK

- Probability & Statistics
- Advanced topics in Networking
- Computer & Network Security
- Network & Web Application Design

REFERENCES

Mr. Chaminda Hettigoda
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ADDITIONAL INFORMATION

- Github : https://github.com/ishann211
- Hackerrank : https://www.hackerrank.com/profile/e20211